A ODD+D Protocol for the Type III Neoclassical Realist Alliance-Network ABM

A.1 Purpose and patterns

The purpose of the model is to examine how cross-country variation in strategic environment, systemic clarity, and material capacity affects state-level alliance orientations and the resulting structure of an international alliance network. The model operationalizes the Type III Neoclassical Realist (NCR) proposition that systemic stimuli are filtered through unit-level conditions, translated into foreign-policy orientations, and aggregated into system-level outcomes.

The model is not designed to predict every formal alliance tie perfectly. Instead, it provides a historically anchored computational environment in which the effects of the three focal systemic variables can be isolated through a neutralized ablation design. In the baseline condition, the observed variation in strategic environment, systemic clarity, and material capacity is retained. In each ablation condition, the variation of one focal input is neutralized by replacing it with its global sample mean. Differences between the baseline and the relevant ablation condition are interpreted as evidence of that input's relative importance within the specified decision mechanism.

The model is designed to reproduce and analyse four broad patterns:

1. the distribution of country-level policy orientations across *Status Quo*, *Balance*, and *Bandwagon*;
2. the number and type of proposed and committed alliance changes;
3. alliance-network edge count, density, modularity, and community structure; and
4. the similarity between the model-generated network and the observed Correlates of War (COW) alliance network in the following year.

Modularity is interpreted as an indicator of alliance-bloc or community structure rather than as a standalone normative measure of political cohesion.

A.2 Entities, state variables, and scales

A.2.1 Entities

The model contains two substantive entity types.

1. **Country agents.** Each sampled country is represented by one `CountryAgent`. The final sample contains 38 countries, selected because their country-year observations are jointly available across the datasets used to construct the independent and intervening variables.
2. **Alliance network.** The international alliance system is represented as an undirected graph. Countries are nodes and formal COW alliance ties are edges. An observed alliance tie is included when its alliance-strength score is at least one. Newly committed balancing ties are stored with strength four, while newly committed bandwagoning ties are stored with strength two.

The model is implemented in Python. Mesa is used for the model and agent classes, NetworkX is used for the alliance graph, and Mesa's `DataCollector` records agent-level and model-level outputs.

A.2.2 Country-agent state variables

At the beginning of each annual step, country agent i holds or receives the following state variables:

- `iso3`: the country's three-letter ISO code;
- `policy`: one of `Status_Quo`, `Balance`, or `Bandwagon`;
- the modified strategic-environment score, `Strategic_Environment_Index_Modified`, and its normalized value;
- the modified clarity score, `Clarity_Index_Modified`, and its normalized value;
- own material capacity, measured through CINC, together with the normalized capacity diagnostic value;
- `primary_threat_iso3`: the country that produces the maximum directed dyadic pressure;
- `primary_threat_pressure` and `threat_capacity`;
- `own_capacity_effective`, `ally_capacity`, `coalition_capacity`, `relative_resistance`, and `vulnerability_score`;
- `action_pressure`, `u_balance`, `u_bandwagon`, `decision_gap`, and `effective_decision_margin`;
- `formalization_allowed`, `best_resistance_gain`, pending alliance additions, and pending alliance removals.

A.2.3 Model-level state variables

The model stores the country-year input table, the directed dyadic table, the historical COW alliance table, the current-year alliance graph, a country-to-agent lookup table, annual capacity maps, the historical formalization-budget table, and validation diagnostics.

The main model-level outputs are:

- candidate and committed alliance additions and removals;
- predicted and observed edge counts;
- edge precision, recall, F1 score, Jaccard similarity, and balanced accuracy;
- predicted and observed network density;
- predicted and observed modularity;
- absolute modularity error;
- normalized mutual information (NMI) and adjusted Rand index (ARI) for community memberships; and
- the annual historical systemic-conflict context indicator.

A.2.4 Spatial and temporal scales

The model has no geographic grid. Geography enters upstream through the dyadic influence weights used to construct strategic pressure. Within the ABM, the interaction space is an undirected alliance network in which each sampled country occupies one node and may be connected to any other sampled country by a formal alliance edge.

Time is discrete. One model step represents one calendar year. The simulation covers 1995–2012 and therefore contains 18 annual rounds. At the beginning of every annual round, the alliance network is reset to the observed COW alliance graph in year t. The model then generates a limited set of theoretically motivated alliance changes and compares the resulting graph with the observed COW network in year $t+1$, where that comparison is available.

The model should therefore be understood as a sequence of historically anchored annual transition experiments. It is not an unrestricted recursive simulation in which the predicted network in year t becomes the starting network in year $t+1$.

A.3 Process overview and scheduling

All country agents are processed synchronously. Their decisions are based on the same observed alliance network and the same country-year information set. Agents do not observe or react to alliance changes proposed by other agents until the model-level formalization stage.

For each annual round t, the model executes the following sequence:

1. Reset the alliance graph to the observed COW alliance network in year t.
2. Load country-year and dyadic values for year t and update the annual capacity maps.
3. Each country agent identifies its primary threat, calculates action pressure and relative resistance, and selects *Status Quo*, *Balance*, or *Bandwagon*.
4. Each country agent stages potential alliance additions and removals, conditional on its selected policy and the formalization gate.
5. The model aggregates all proposed changes, applies the receiver-acceptance rule, estimates the historical formalization budget, ranks candidate changes, and commits only the highest-scoring changes within that budget.
6. The model records agent-level and system-level outputs.
7. If data for year $t+1$ are available, the model compares the generated network with the observed COW network in year $t+1$.
8. The model advances to the next year and repeats the process until 2012.

A.4 Design concepts

A.4.1 Basic principles

The model implements a Type III NCR mechanism. Strategic environment represents the intensity of systemic pressure. Systemic clarity represents how legible that pressure is to the state. Material capacity determines the extent to which a state can resist its most salient threat, either independently or with support from existing allies.

The unit-level intervening variables are incorporated upstream during data construction. They modify the country-year strategic-environment and clarity inputs before those inputs enter the ABM. The model therefore does not treat the domestic filters as independently acting agents. Instead, it uses their already modified country-year effects in the state-level decision rule.

A.4.2 Emergence

The principal emergent outcome is the structure of the model-generated alliance network. Individual country decisions can generate variation in edge count, density, modularity, community memberships, and similarity to the observed COW network.

No country is assigned to a bloc or community before the simulation. Communities are identified from the final alliance network through greedy modularity optimization.

A.4.3 Adaptation

Agents do not learn, imitate, or revise their behavioural rules over time. Their behaviour changes only because country-year inputs, primary threats, observed alliance ties, and available partners differ across annual rounds.

A balancing agent evaluates potential partners using the current alliance network and current annual capacity map.

A.4.4 Objectives

Agents do not solve a global optimization problem. Their local objective is represented by a rule-based comparison between balancing utility and bandwagoning utility after systemic pressure exceeds the threshold required for action.

Balancing is favoured when relative resistance against the primary threat is sufficiently high. Bandwagoning is favoured when vulnerability is sufficiently high. When the difference between the two options is small, especially under low clarity, the agent retains the status quo.

A.4.5 Learning

There is no learning mechanism. Country agents do not retain memory of their previous model-generated policy choices. This is intentional because each annual round begins from the observed alliance network in that year.

A.4.6 Prediction

The model does not claim to provide exact forecasts of individual alliance ties. It performs one-year-ahead structural comparisons whenever the observed COW network in year $t+1$ is available. These comparisons function as external plausibility and structural-validation diagnostics rather than as the dependent variable of a forecasting exercise.

A.4.7 Sensing

Each country agent observes:

- its own modified strategic-environment and clarity scores;
- its own material capacity;

- the directed dyadic pressures generated by all other sampled countries;
- the capacities of current allies and potential partners; and
- the observed alliance network in the current year.

The agent identifies one primary threat: the sender country that generates the maximum directed dyadic pressure.

A.4.8 Interaction

Interaction occurs in two forms. First, countries interact indirectly through the directed dyadic pressure structure used to identify primary threats. Second, countries interact through the undirected alliance network.

A balancing agent avoids a potential partner that is already allied with its primary threat. A proposed alliance tie can be rejected when the intended recipient is balancing against the proposing country. All proposed changes are resolved during the model-level formalization stage.

A.4.9 Stochasticity

The current implementation contains no stochastic behavioural draw, random partner-matching procedure, or random activation order. A fixed seed of 616 is retained for reproducibility, but the implemented decision rules are deterministic. Given identical data and parameter values, the model produces identical results.

Variation across reported conditions therefore comes from the neutralized ablation design rather than from Monte Carlo randomness. Any future extension that introduces stochastic choice, random activation, or parameter sampling should report multiple independent replications.

A.4.10 Collectives

The alliance graph and the communities detected within it are the model's collectives. Collective structure is measured through modularity and community labels. The model does not assign collective preferences or group-level decision rules.

A.4.11 Observation

The `DataCollector` records country-year values for policy orientation, primary threat, normalized decision inputs, resistance, vulnerability, action pressure, utilities, formalization eligibility, and proposed changes.

At the model level, it records candidate and committed changes, predicted and observed edges, edge precision, recall, F1 score, Jaccard similarity, balanced accuracy, density, modularity, modularity error, community NMI, community ARI, and the historical systemic-conflict indicator.

A.5 Decision-making of `CountryAgent` (ODD+D extension)

A.5.1 Theoretical and empirical background

Although the model's agents represent states rather than literal individuals, the policy rule abstracts foreign-policy choices made through state decision makers. The decision mechanism follows the Type III NCR distinction between systemic stimuli and the domestic or perceptual filters through which states interpret those stimuli.

Strategic environment captures the magnitude of externally relevant pressure. Systemic clarity affects whether that pressure is sufficiently interpretable to trigger action. Material capacity, including support from existing allies, determines whether a state can plausibly resist its primary threat.

States with sufficient action pressure and relatively high resistance select balancing. States with sufficient action pressure but high vulnerability select bandwagoning. Ambiguous cases retain the status quo.

A.5.2 Information inputs and primary-threat identification

For country i in year t, the agent receives modified strategic environment $S_{i,t}$, modified systemic clarity $C_{i,t}$, own capacity $\mathrm{CINC}_{i,t}$, directed dyadic influence weights $w_{ij,t}$, the capacities of other countries $\mathrm{CINC}_{j,t}$, and the observed alliance network G_t.

Directed dyadic pressure is defined as:

$$P_{ij,t} = w_{ij,t} \times \mathrm{CINC}_{j,t}. \tag{1}$$

The primary threat is the state producing the largest directed pressure:

$$j^{\star}_{i,t} = \arg\max_{j \neq i} \left(P_{ij,t}\right). \tag{2}$$

This rule means that the primary threat is not necessarily the geographically closest, most hostile, or most capable country in isolation. It is the country that combines strategic relevance, as captured by $w_{ij,t}$, with material capacity.

A.5.3 Action pressure

The modified strategic-environment and clarity inputs are min–max normalized across the full country-year model sample to the interval $[0, 1]$. Let the normalized values be denoted by $S_{i,t}$ and $C_{i,t}$.

Action pressure is defined as:

$$A_{i,t} = S_{i,t}\left[(1-\alpha) + \alpha C_{i,t}\right], \qquad \alpha = 0.50. \tag{3}$$

Clarity does not independently create action pressure. Instead, it determines how strongly strategic-environment pressure is translated into an actionable signal. The agent remains in the status quo whenever:

$$A_{i,t} < 0.08. \tag{4}$$

The normalized capacity score is recorded for diagnostic and ablation purposes. However, capacity affects the substantive policy decision primarily through the raw CINC values used in the relative-resistance calculation below.

A.5.4 Relative resistance and vulnerability

Let $\mathcal{A}_{i,t}$ denote the set of current alliance partners of country i in year t. The model excludes the primary threat from the ally-capacity calculation even when the threat is an existing alliance partner.

Coalition capacity is:

$$K_{i,t} = \mathrm{CINC}_{i,t} + \sum_{a \in \mathcal{A}_{i,t} \setminus \{j^{\star}_{i,t}\}} \mathrm{CINC}_{a,t}. \tag{5}$$

Relative resistance is then:

$$R_{i,t} = \frac{K_{i,t}}{K_{i,t} + \mathrm{CINC}_{j^{\star}_{i,t},t}}. \tag{6}$$

Vulnerability is defined as:

$$V_{i,t} = 1 - R_{i,t}. \tag{7}$$

A high value of $R_{i,t}$ indicates that country i can more plausibly resist its primary threat. A high value of $V_{i,t}$ indicates that accommodation with that threat is relatively more attractive.

A.5.5 Policy selection

The model computes balancing and bandwagoning utilities as:

$$U^{\mathrm{Balance}}_{i,t} = A_{i,t} R_{i,t}, \tag{8}$$

$$U^{\mathrm{Bandwagon}}_{i,t} = A_{i,t} V_{i,t}. \tag{9}$$

The decision gap is:

$$D_{i,t} = U^{\mathrm{Balance}}_{i,t} - U^{\mathrm{Bandwagon}}_{i,t}. \tag{10}$$

The effective decision margin widens when systemic clarity declines:

$$m_{i,t} = 0.02 + 0.08(1 - C_{i,t}). \tag{11}$$

The policy rule is:

$$\begin{cases} \text{Status Quo}, & \text{if } A_{i,t} < 0.08, \\ \text{Balance}, & \text{if } D_{i,t} > m_{i,t}, \\ \text{Bandwagon}, & \text{if } D_{i,t} < -m_{i,t}, \\ \text{Status Quo}, & \text{if } |D_{i,t}| \leq m_{i,t}. \end{cases} \tag{12}$$

The uncertainty region is substantively important. It prevents agents from treating small utility differences as decisive when the international environment is unclear.

A.5.6 From policy orientation to proposed formal alliance ties

Policy orientation is not equivalent to immediate formal alliance change. A country must pass a second formalization gate. It must select either balancing or bandwagoning and satisfy:

$$A_{i,t} \geq 0.12. \tag{13}$$

A bandwagoning country proposes a tie to its primary threat. A newly committed bandwagoning tie is stored with alliance strength two.

A balancing country first stages the removal of an existing alliance tie to its primary threat, where such a tie exists. It then searches for a potential partner k that satisfies all of the following conditions:

- k is not country i itself;
- k is not the primary threat $j^{\star}_{i,t}$;
- k is not already allied with country i;
- k is not already allied with country i's primary threat.

For every eligible candidate, the model calculates expected resistance after adding that candidate, $R^{(+k)}_{i,t}$. Candidate selection is based on:

$$Q_{ik,t} = \max\left(0, R^{(+k)}_{i,t} - R_{i,t}\right) + 0.10 \cdot \mathbb{I}\left(j^{\star}_{k,t} = j^{\star}_{i,t}\right), \tag{14}$$

where $\mathbb{I}(\cdot)$ equals one when the candidate identifies the same primary threat and zero otherwise.

The selected balancing partner is the eligible candidate with the highest $Q_{ik,t}$. A newly committed balancing tie is stored with alliance strength four.

A.5.7 Acceptance and formalization

The model-level formalization process applies two constraints.

First, a receiving country rejects a proposed alliance tie when it is balancing against the proposer. More specifically, a proposal from country i to country k is rejected when country k has selected *Balance* and identifies country i as its primary threat.

Second, the model limits the number of alliance changes that can become formal ties in each year. The historical formalization budget is:

$$B_t = \text{round}\left(\frac{1}{L_t}\sum_{\ell\in\mathcal{L}_t} \text{ObservedChanges}_{\ell}\right), \tag{15}$$

where $\mathcal{L}_t$ contains up to the five available years preceding t, L_t is the number of available years in that lookback window, and $\text{ObservedChanges}_{\ell}$ is the total number of observed COW alliance additions and removals between year ℓ and the following year. When no previous historical information is available, the budget is zero.

All surviving candidate additions and removals are ranked by score. For additions, the ranking score is the relevant policy utility plus the balancing resistance gain where applicable. For removals, the ranking score is the proposing country's action pressure. The model commits only the top B_t changes.

The resulting graph is the annual model-generated alliance network:

$$\widehat{G}_t. \tag{16}$$

A.6 Details

A.6.1 Initialization

At initialization, the model reads the prepared country-year, directed dyadic, and historical alliance datasets. It creates one graph node and one `CountryAgent` for each sampled country. The initial graph is the observed COW alliance graph in 1995.

However, the graph is reset to the observed COW graph at the beginning of every subsequent annual round. Initialization therefore does not establish an endogenous long-run path dependence across model-generated graphs.

Table 1: Baseline model parameters.

Parameter	**Value**	**Role**
Start year	1995	First annual simulation round.
Final year	2012	Final annual simulation round.
Number of countries	38	Countries with joint coverage across all required inputs.
Clarity action weight, α	0.50	Weight of clarity in the action-pressure function.
Threat activation threshold	0.08	Minimum action pressure required to leave the status quo.
Base decision margin	0.02	Minimum utility separation under full clarity.
Uncertainty margin	0.08	Additional decision margin under low clarity.
Formalization threshold	0.12	Minimum action pressure required to stage formal alliance change.
Bandwagoning tie strength	2	Stored strength of a newly committed bandwagoning tie.
Balancing tie strength	4	Stored strength of a newly committed balancing tie.
Shared-threat bonus	0.10	Bonus applied when a balancing candidate identifies the same primary threat.
Formalization lookback	5 years	Historical COW window used to estimate the annual change budget.
Minimum observed alliance strength	1	Threshold required for a COW alliance tie to be included in the graph.
Seed	616	Reproducibility setting; no active stochastic behavioural process exists in the current implementation.

A.6.2 Input data

The model uses three prepared inputs.

1. **Country-year data.** This table contains the modified strategic-environment and clarity indices, produced after incorporating the intervening-variable filters described in the Research Design. It also includes country and year identifiers needed to link observations to agents.

2. **Directed dyadic data.** This table contains $w_{ij,t}$, both countries' annual CINC values, dyadic pressure, and annual hostility information. Dyadic pressure is calculated as $w_{ij,t} \times \text{CINC}_{j,t}$.

3. **Alliance data.** This table contains annual COW formal-alliance records and their alliance-strength values. It is used both to reconstruct the observed starting graph in every year and to estimate the historical formalization budget.

The ABM does not independently estimate strategic environment, systemic clarity, or the domestic-filter indices. It consumes the prepared country-year and dyadic values generated by the empirical data-construction pipeline.

A.6.3 Observed-network anchoring

At the start of each annual round, the model clears all existing graph edges and reconstructs the observed COW alliance network for the current year among the sampled countries.

This procedure avoids the unrealistic implication that every alliance tie must emerge from an empty graph. It also means that model-generated changes should be interpreted as locally generated departures from a historically observed alliance structure rather than as a complete independent reconstruction of world alliance history.

A.6.4 Agent observation and policy decision

Each country agent retrieves its country-year values and all directed dyads in which it is the receiving state. It identifies the primary threat, calculates action pressure, relative resistance, vulnerability, balancing utility, bandwagoning utility, and the effective decision margin, and then selects one of the three policy orientations according to Equation 12.

A.6.5 Alliance-change staging

Each country agent translates a non-status-quo policy into at most one proposed addition and, where relevant, one proposed removal. The staging phase does not immediately change the network.

This separation preserves synchronous updating because all country agents calculate their policy choices from the same observed network G_t.

A.6.6 Formalization and graph updating

The model gathers all staged proposals, applies the recipient-acceptance rule, ranks candidate additions and removals, and commits only the highest-scoring changes within the annual formalization budget.

The resulting graph $\widehat{G}_t$ captures the combination of policy preferences and the institutional stickiness of formal alliances. A state may prefer a different alignment without being able to formalize that preference immediately.

A.6.7 Outcome measurement and validation diagnostics

For each year, the model records country-level mechanisms and system-level network outcomes. The model-generated graph is compared with the observed COW graph in year $t + 1$ whenever data for the target year are available.

Edge-level comparison includes precision, recall, F1 score, Jaccard similarity, and balanced accuracy. Structural comparison includes predicted and observed density, predicted and observed modularity, absolute modularity error, normalized mutual information (NMI), and adjusted Rand index (ARI) for community labels.

The annual systemic-conflict indicator is calculated from COW Militarized Interstate Dispute hostility data as the sum of directed maximum hostility values among sampled countries divided by two. It is retained as historical context and is not an endogenous conflict process generated by the ABM.

A.6.8 Neutralized ablation experiments

The model runs four conditions:

1. **Baseline:** observed variation in all focal inputs is retained.
2. **Neutralized strategic environment:** the normalized strategic-environment score used in action pressure is replaced by its global sample mean.
3. **Neutralized systemic clarity:** the normalized clarity score used in action pressure and the uncertainty margin is replaced by its global sample mean.
4. **Neutralized capacity:** raw CINC values used for own capacity, ally capacity, candidate capacity, and threat capacity are replaced by the global mean raw CINC value.

In all three ablation conditions, the observed dyadic pressure map remains available for identifying the primary threat. Consequently, neutralization isolates the role of variation in the focal decision input without eliminating the historically grounded dyadic context through which threats are identified.

For each neutralized variable x, the analysis calculates policy deviation, mechanism deviation, and system deviation relative to the baseline. The overall determinacy score is:

$$\text{Determinacy}_x = \frac{\text{PolicyDeviation}_x + \text{MechanismDeviation}_x + \text{SystemDeviation}_x}{3}. \tag{17}$$

The focal variables are ranked according to their overall determinacy scores. A higher score indicates that neutralizing the corresponding variable produces a larger deviation from the baseline model within the specified mechanism.

A.7 Reproducibility and reporting note

This ODD+D specification describes the current implementation of the model. The manuscript should make explicit that:

- the alliance network is re-anchored to observed COW data at the beginning of every year;
- country decisions are synchronous;
- formal alliance changes are constrained by an empirical historical change budget;
- the current implementation is deterministic; and
- the model is designed to examine theoretically specified mechanisms and structurally plausible outcomes rather than exact alliance prediction.